Supply Chain

R3-Data Analyst Specialist

CAI3_DAT1_G1

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Introduction

Definition of Supply Chain:

Supply chain is the process of getting a product or service from its origin to the final consumer. It includes the network of all the organizations, people, activities, information, and resources involved.

The main goal of supply chain is to ***deliver the correct orders to right place on time with no errors or returns with low costs to get higher customer satisfaction***.

Steps of supply chain:

- **Sourcing raw materials**: the procurement of raw materials and components
- **Manufacturing**: the process of turning raw materials into finished goods
- **Logistics and delivery**: managing storage (warehousing), transportation, order processing, and the final delivery to the customer
- **Returns**: the process of handling customer returns

Most Popular KPIs of supply chain:

1. Cash to Cash Cycle Time

It represents the average length of time required to convert resources into cash flows, starting from the moment you pay for inventory to the time you collect money for the sale of that inventory. To measure it, three common financial metrics are used namely: days of inventory (DOI), days of payables (DOP), and days sales outstanding (DSO). Add DOI and DOP, then subtract DSO to arrive at cash-to-cash cycle time.

2. Perfect Order Rate

The perfect order rate represents the percentage of orders that are delivered in full, on time, without incident, and with documentation that is accurate and complete. It directly impacts customer satisfaction.

3. Fill Rate

It represents the percentage of orders that are successfully completed with the first shipment and do not require a second, third, or fourth shipment, and so on. It also impacts on customer satisfaction.

4. Customer Order Cycle Time

It refers to the average amount of time (in days) that lapses between the date the customer places an order and the actual delivery date. It reflects the management of receivables and inventory.

5. Inventory Turnover Rate

It measures how often the inventory is sold and replaced within a specific timeframe, such as a month or a year. It reflects how efficiently a company is managing its stock. It can be calculated by dividing the cost of goods sold by the average inventory for a specific period. To find the average inventory, add the beginning inventory value to the ending inventory value and divide the sum by two. The resulting number indicates how many times inventory was sold and replenished during that period.

There are many other indicators that can efficiently evaluate the supply chain performance such as: **On-Time delivery** – **On-Time shipment** and **Reasons for return**.

Main Steps:

1- **Domain Knowledge**

2- **Data cleaning**

3- **Normalization**

4- **Analysis Questions**

5- **Visualization**

Conclusion and Recommendation

Domain Knowledge

1-Domain Knowledge:

In this step, we depended on 2 main sources:

1- **Searching and using AI:**

To understand the business context and the dataset. Identifying key variables such as inventory levels, order quantities, lead times, supplier performance, costs, revenue.

2- **Many discussions with Instructor:**

To guide us to fill gaps in data to highlight the main KPIs (key analytical questions) that can be extracted (meaningful insights)

Main indicators in Dataset:

100 Rows (Products)	3 Product Types		
	Cosmetics (26) - Skincare (40) - Haircare (34)		
Cities (5) - India Orders:	Mumbai – Kolkata – Delhi – Bangalore - Chennai (22) – (25) – (15) – (18) – (20)		
No of Orders	100		
Total Order Quantities	4922		
No of Suppliers	5	(1 – 2 – 3 – 4 - 5)	
No of Carriers	3	(A – B - C)	
Transportation Modes	4	(Air – Sea – Rail - Road)	
No of Routes	3	(A – B - C)	
		23 Variables	
Main Variables	**Min**	**Max**	**Average**
Stock %	0	100	47.7
Availability %	1	100	48.4
Total Costs	103٫9 $	997٫4 $	529.3 $
Total Revenue	1061٫6 $	9866٫5 $	5776٫1
Total Lead Time	19 d	90 d	54 d
Defect Rate	0.02	4.9	2.3
Inspection Result	Pass: 23%	Fail: 36%	Pending: 41%

Data Cleaning

2-Data Cleaning

First: using Python

- **Read file in python**

jupyter Project Last Checkpoint: 4 minutes ago

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```
[1]: import numpy as np
     import pandas as pd
```

```
[2]: df=pd.read_csv('supply_chain_data.org.csv')
```

```
[3]: df
```

[3]:

	Product type	SKU	Price	Availability	Number of products sold	Revenue generated	Customer demographics	Stock levels	Lead times	Order quantities	...	Location	Lead time	Production volumes	Manufacturing lead time	Manufa
0	haircare	SKU0	69.808006	55	802	8661.996792	Non-binary	58	7	96	...	Mumbai	29	215	29	46.
1	skincare	SKU1	14.843523	95	736	7460.900065	Female	53	30	37	...	Mumbai	23	517	30	33.
2	haircare	SKU2	11.319683	34	8	9577.749626	Unknown	1	10	88	...	Mumbai	12	971	27	30.
3	skincare	SKU3	61.163343	68	83	7766.836426	Non-binary	23	13	59	...	Kolkata	24	937	18	35.
4	skincare	SKU4	4.805496	26	871	2686.505152	Non-binary	5	3	56	...	Delhi	5	414	3	92.
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

- **Checking missing values:**

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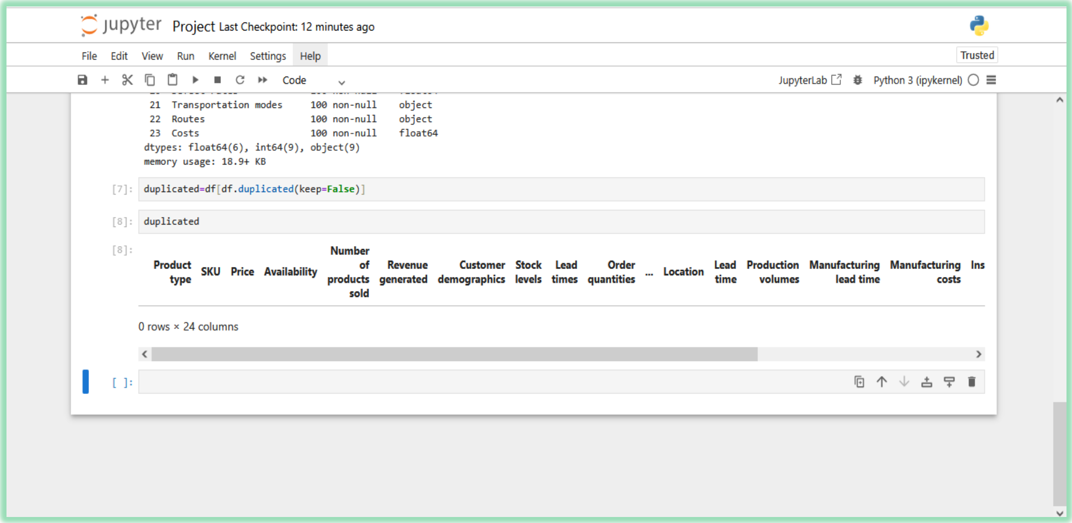
```
[3]: df.isnull().sum()
```

```
[3]: Product type                0
     SKU                         0
     Price                       0
     Availability                0
     Number of products sold     0
     Revenue generated           0
     Customer demographics       0
     Stock levels                0
     Lead times                  0
     Order quantities            0
     Shipping times              0
     Shipping carriers           0
     Shipping costs              0
     Supplier name               0
     Location                    0
     Lead time                   0
     Production volumes          0
     Manufacturing lead time     0
     Manufacturing costs         0
     Inspection results          0
     Defect rates                0
     Transportation modes        0
     Routes                      0
     Costs                       0
     dtype: int64
```

There are no missing values in all variables in dataset

Also, the data types of all variables are suitable for the data inserted into each one of them

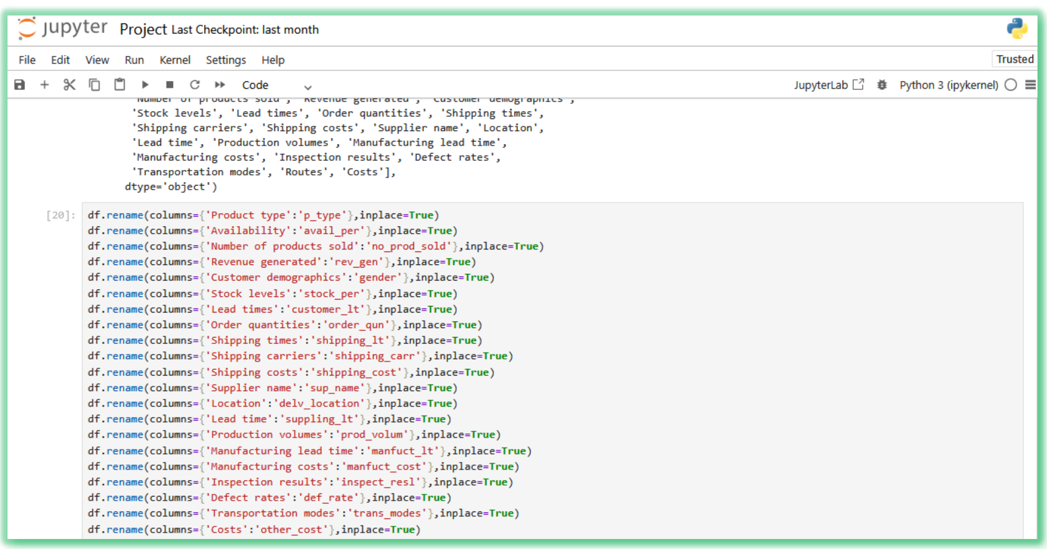
- **Checking duplicated values:**



There are no duplicated values in dataset

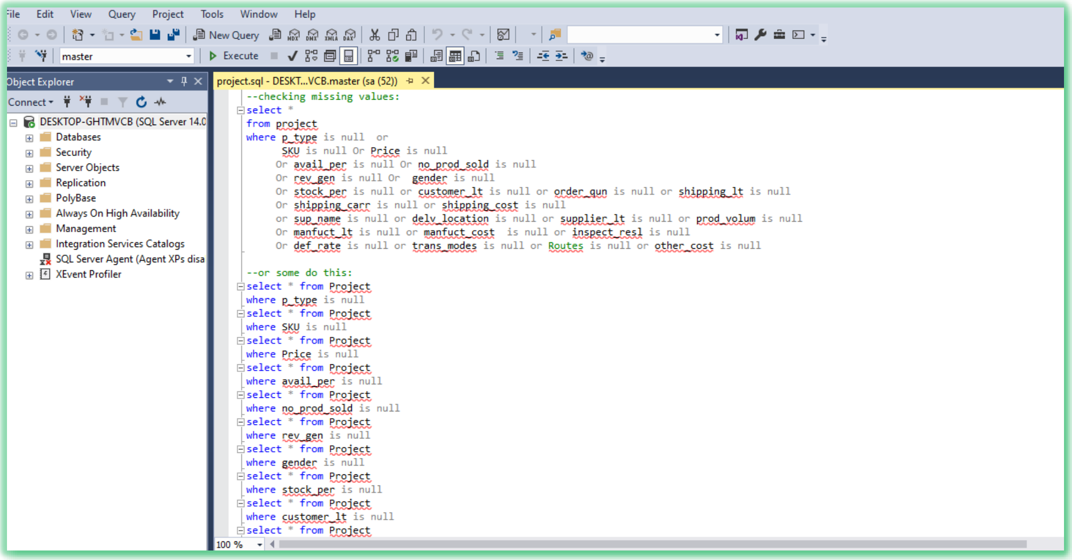
- **Renaming variables:**

  To avoid ambiguous columns names, we renamed many columns



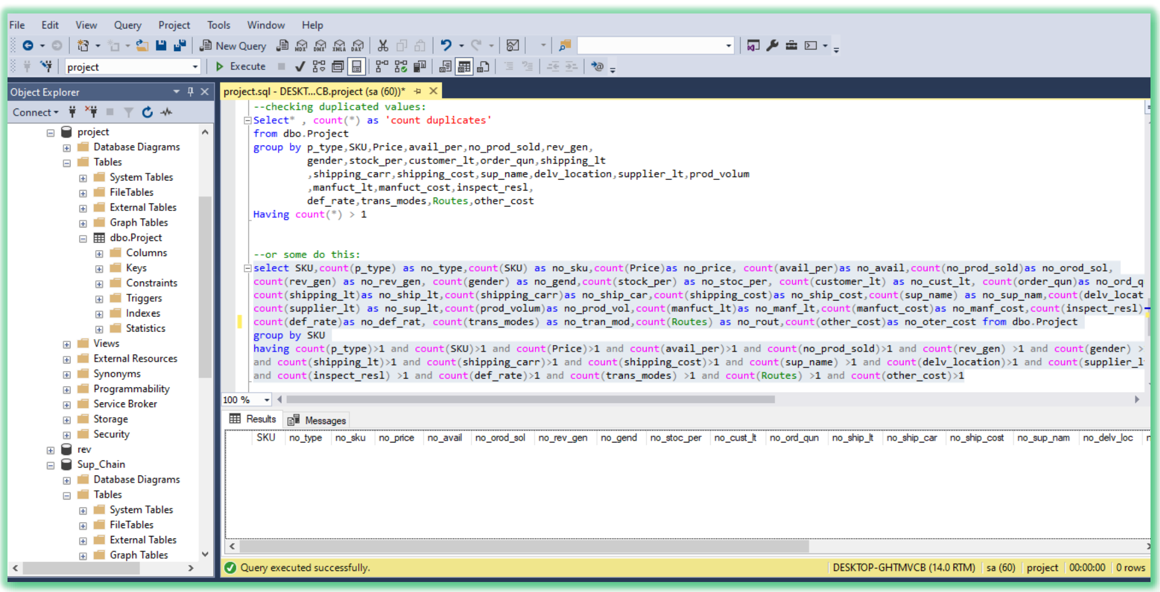
Second: using SQL

- **Checking missing values:**



There are no missing values, so, the result of running this query for all columns (23) is no record

- **Checking duplicated values**



There are no duplicated values, the result of running this query is no records

- **Renaming column:**

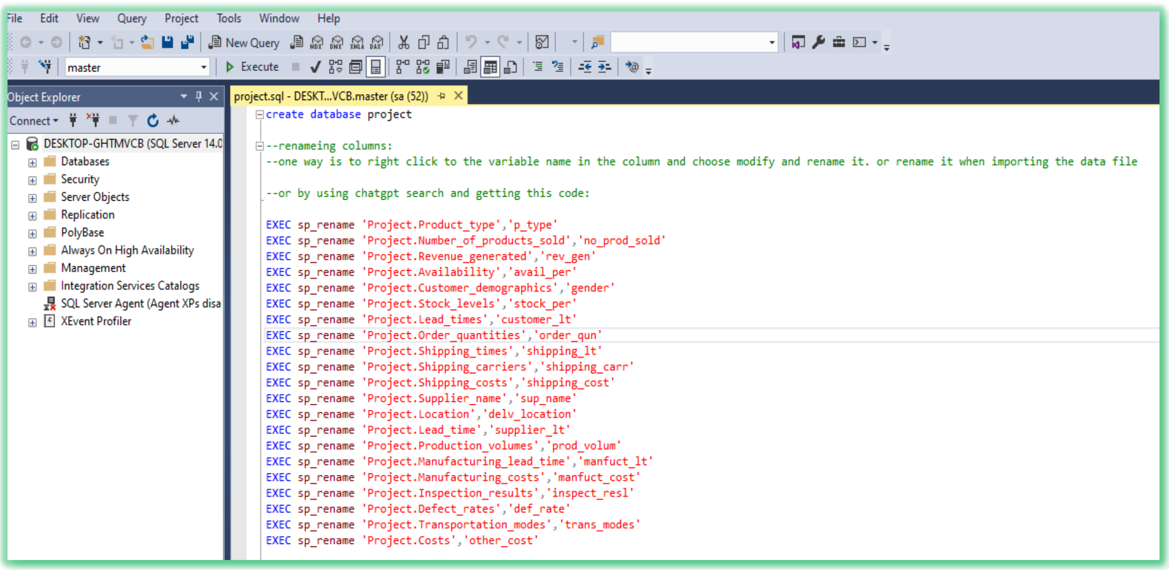
We use this query to rename all columns with ambiguous names. There are 3 methods we can use to rename column in SQL:

1- When importing files to SQL, we can rename columns in importing steps, or

2- We can right click on column name and choose to modify then rename it, or

3- ask ChatGPT and we get this command (renaming) in SQL_Server.

```
exec sp_rename 'project.availablity', 'avail_per'
```

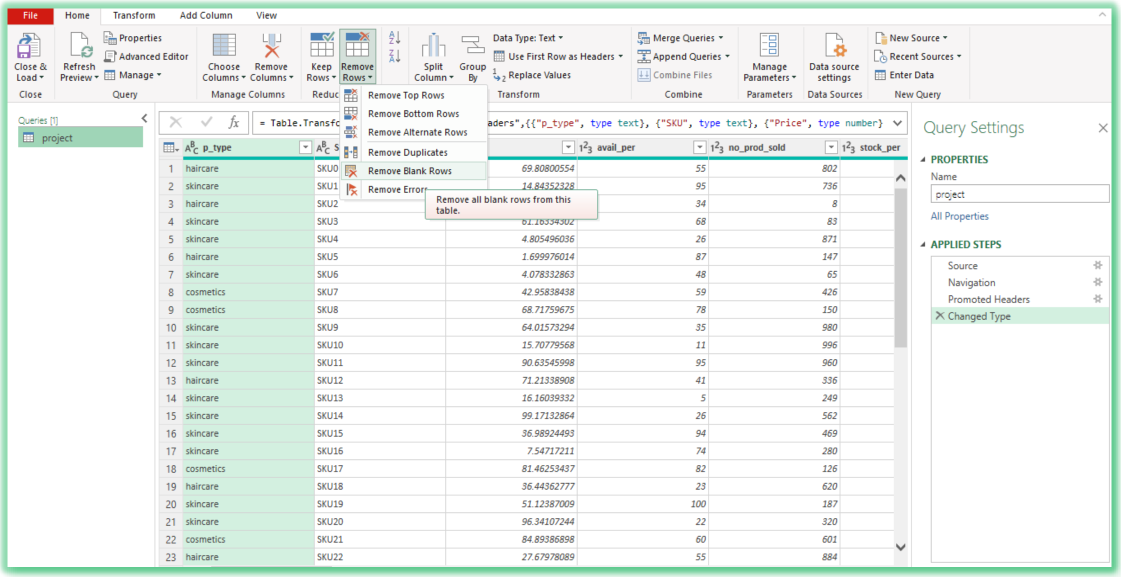


Third: using Excel

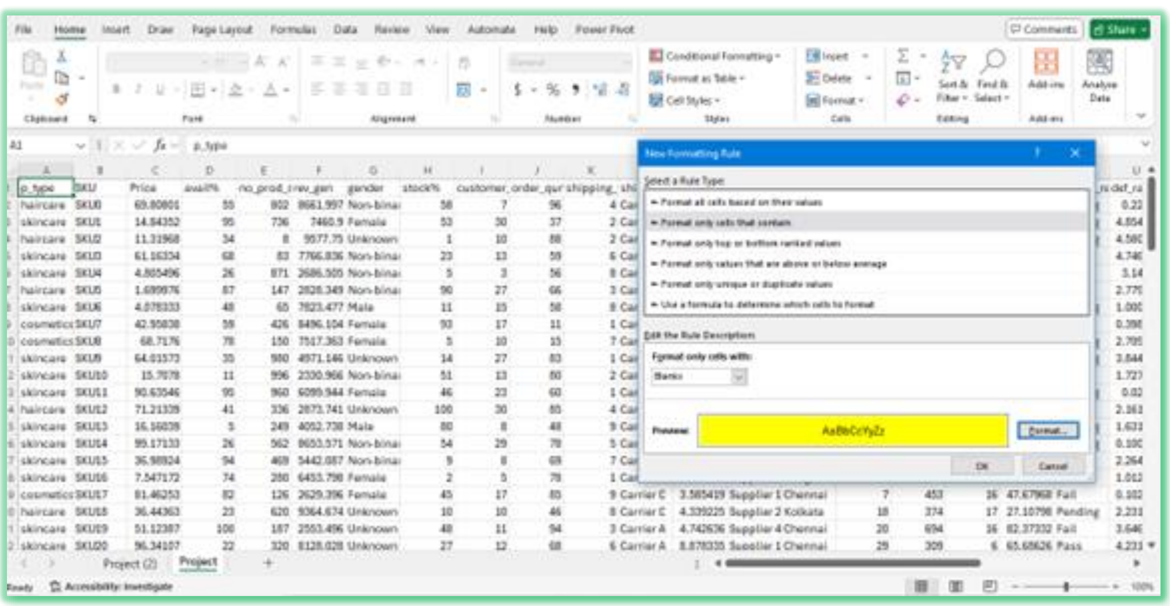
- **checking missing values:**

We can do this in two ways:

1- using power query



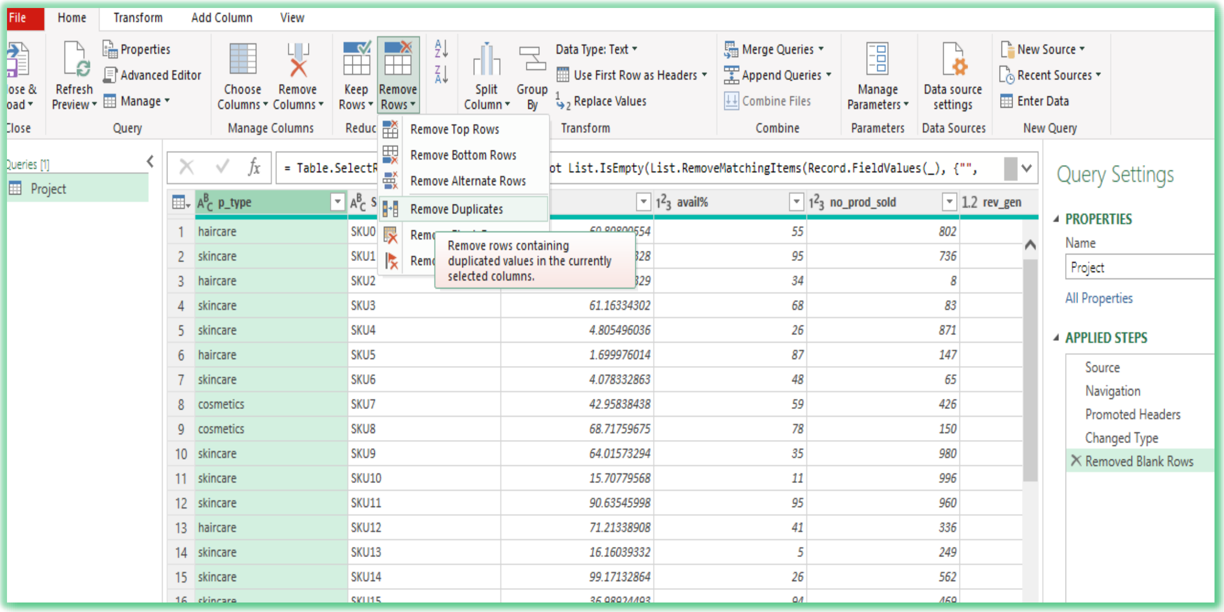
2- or by using conditional formatting to highlight blanks and then remove it:



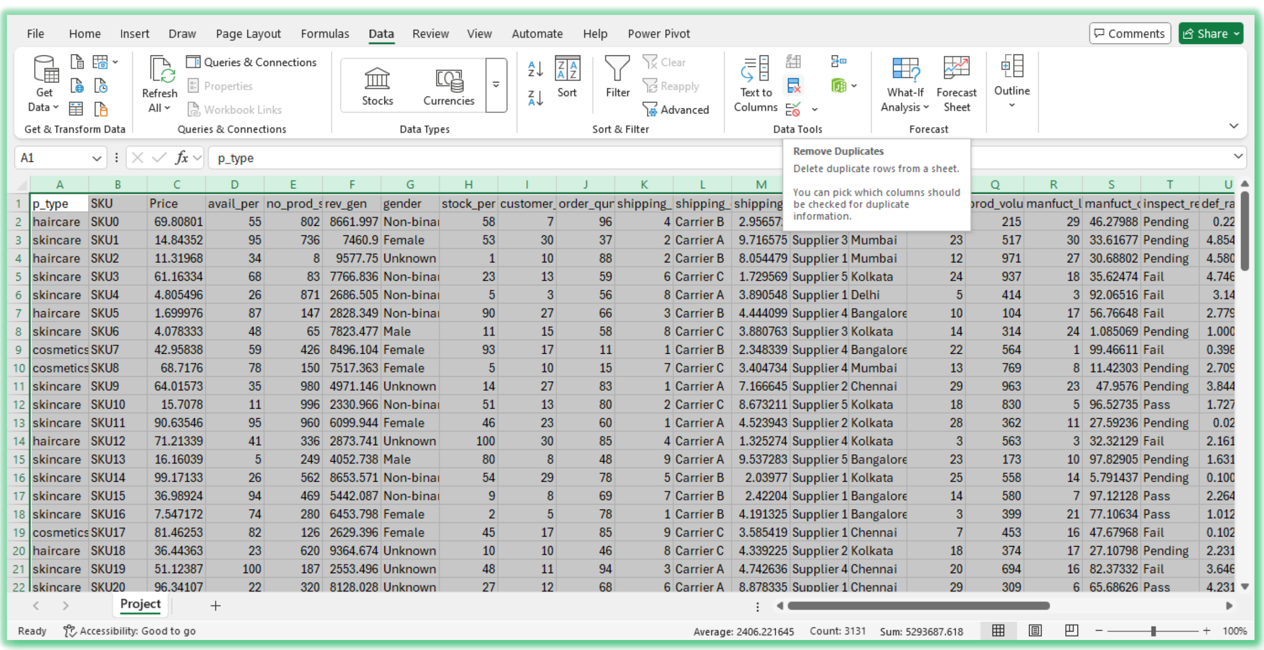
- **Checking duplicated rows:**

  Also, it can be done using three ways:

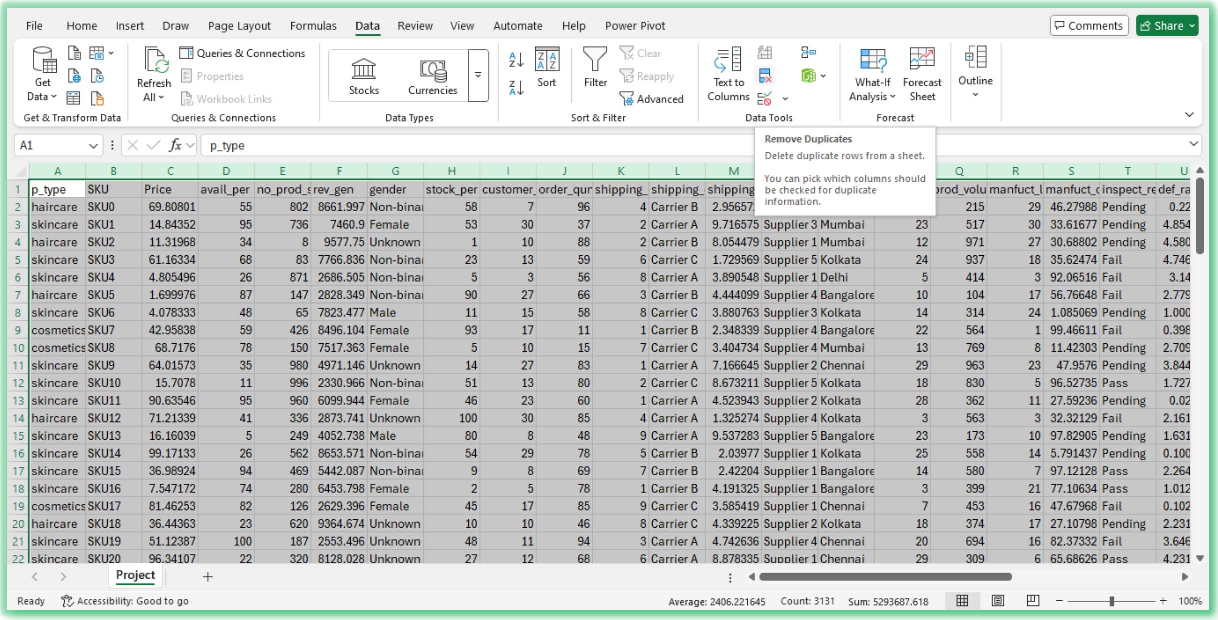
  1- Using power query



  2- Or using conditional formatting to highlight duplicated rows and then remove it.



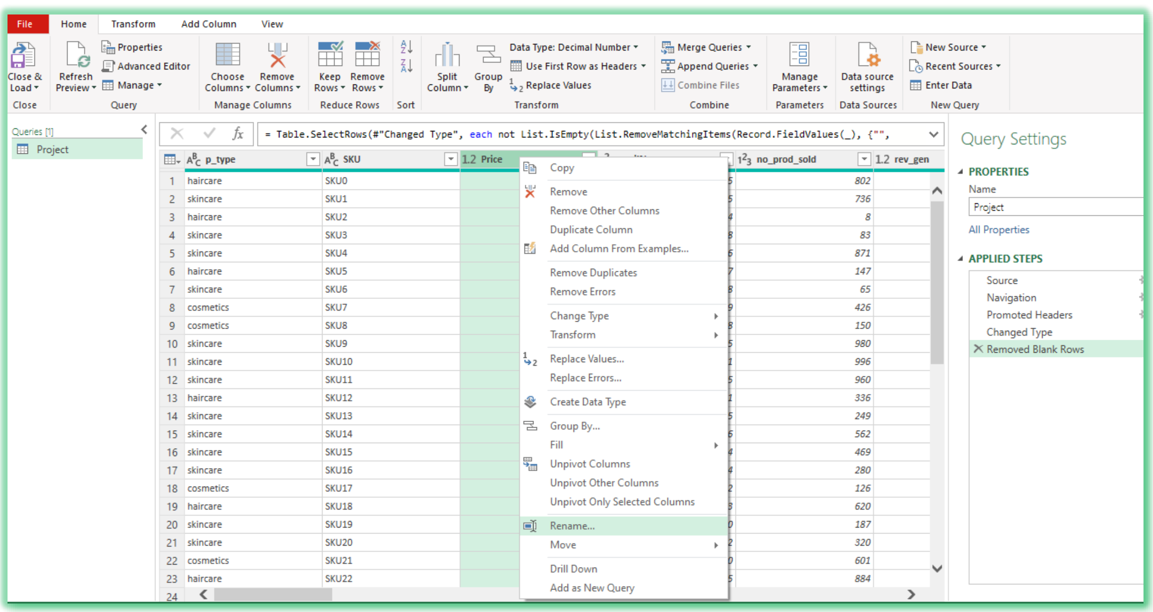
3- Or choose (remove duplicates) from data menu after selecting the whole table



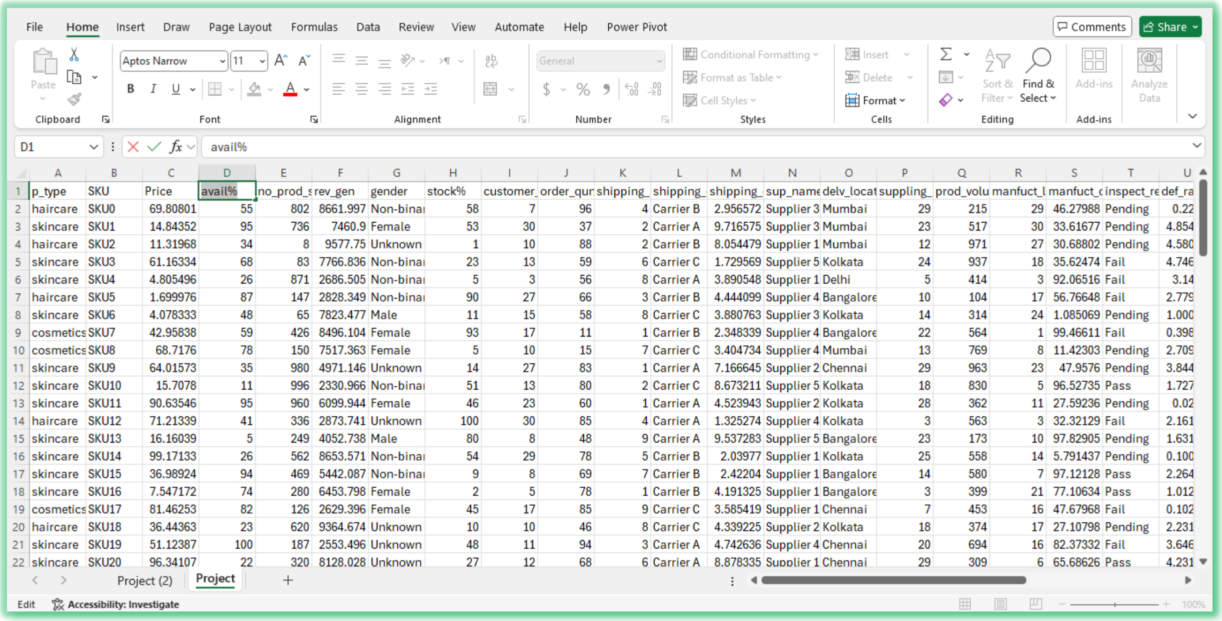
- **Rename columns:**

  Also, there are two ways:

  1- Using power query:



2- In sheets of data, we can just select the column name and change it:



Unknown gender:

In gender variable there are 31 (unknown) cases. We found that males buy **haircare** products more than the other two types (cosmetics and skincare), on contrary, females buy the other two types more. Accordingly, we can assume that the gender of 15 unknown cases whose buy haircare can be males, and unknown cases who buy (cosmetics and skincare) can be females.

Count of SKU	Column Labels				
Row Labels	**Female**	**Male**	**Non-binary**	**Unknown**	**Grand Total**
cosmetics	**10**	4	5	**7**	26
haircare	2	**10**	7	**15**	34
skincare	**13**	7	11	**9**	40
Grand Total	**25**	**21**	**23**	**31**	**100**

The variables names after cleaning

Index	Original Names	Understanding Names (for explanation)	Names for analysis (shortcuts with no spaces – with no special characters)
1	Product type	Category	p_type
2	SKU	product ID	SKU
3	Price	Price	Price
4	Number of products sold	No products sold	no_prod_sold
5	Revenue generated	Revenue generated	rev_gen
6	Availability	Availability% Availability level percentage for SKU = (Units available / Total units demanded) x 100 Available stock (units): The on-hand stock minus reserved quantities	avail_per
7	Customer demographics	Gender	Gender
8	Stock Level	Stock Level% Stock level percentage for SKU = (On-hand stock of that SKU) / (Maximum Capacity of a warehouse for that SKU) x 100	stock_per
9	Lead times	Customer Lead time	customer_lt
10	Order quantities	No of Orders	order_qun
11	Shipping times	Shipping times	shipping_lt
12	Shipping carriers	Shipping carriers	shipping_carr
13	Shipping costs	Shipping costs	shipping_cost
14	Supplier name	Supplier name	sup_name
15	Location	Location	delv_location
16	Lead time	Supplier Lead time	supplier_lt
17	Production volumes	Production volumes	prod_volum
18	Manufacturing lead time	Manufacturing lead time	manfuct_lt
19	Manufacturing costs	Manufacturing costs	manfuct_cost
20	Inspection results	Inspection results	inspect_resl
21	Defect rates	Defect rates	def_rate
22	Transportation modes	Transportation modes	trans_modes
23	Routes	Routes	Routes
24	Costs	total Costs	total_cost

Calculated columns:

Other_costs = total costs – (shipping costs + manufacturing costs)

Total_times = customer lead time + shipping lead time + manufacturing lead time + supplier lead time

Normalization

3- Normalization

SQL normalization is a systematic process for organizing data in a relational database to reduce data redundancy and improve data integrity. It involves decomposing large tables into smaller, related tables and defining relationships between them using primary and foreign keys. The goal is to minimize data anomalies (insertion, update, and deletion anomalies) and ensure data consistency.

Normalization follows a series of rules known as normal forms (NF), each addressing specific types of data redundancy and dependency issues:

First Normal Form (1NF):

- Ensures each column contains atomic (indivisible) values.
- Each row must be uniquely identifiable by a primary key.

Our data set already satisfies the requirements for First Normal Form (1NF):

- Each cell in spreadsheet contains a single, indivisible value (e.g., a single product name, a single quantity, a single supplier ID, etc.).
- Each row is uniquely identifiable by a primary key.

Second Normal Form (2NF):

- Satisfies 1NF.
- Removes partial dependencies, meaning all non-key attributes must be fully dependent on the entire primary key.

Third Normal Form (3NF):

- Satisfies 2NF.
- Eliminates transitive dependencies, meaning non-key attributes should not depend on other non-key attributes.

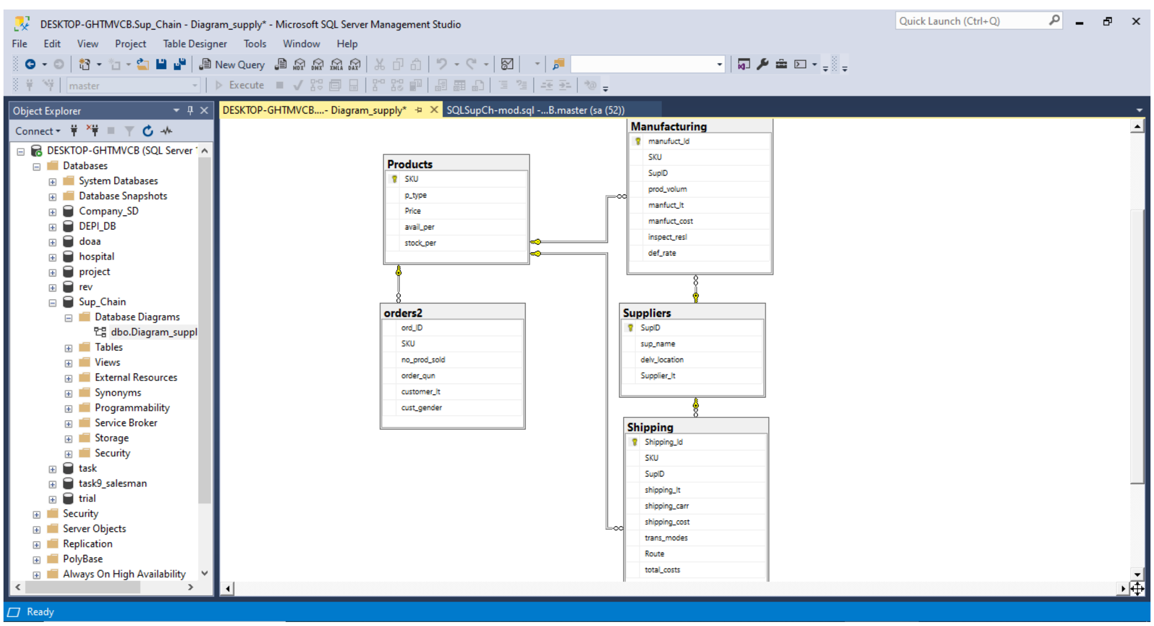
Finally, we have 5 tables as follow:

Products Table: (sku, product type, price, availability_per, stocklecel_per),
Orders Table: (orderID, sku, orders quantities, number of products sold, customer lead time, customer gender),
Suppliers Table: (supplierID, supplierName, Location, supplier lead time),
Manufacturing Table: (manufacturing ID, sku, supplierID, producttion volumes, manufacturing lead time, manufacturing costs, inspection results, defect rates),
Shipping Table: (shippingID, sku, supplierID, shipping lead time, shipping carriers, shipping costs, transportation modes, routes, total costs)

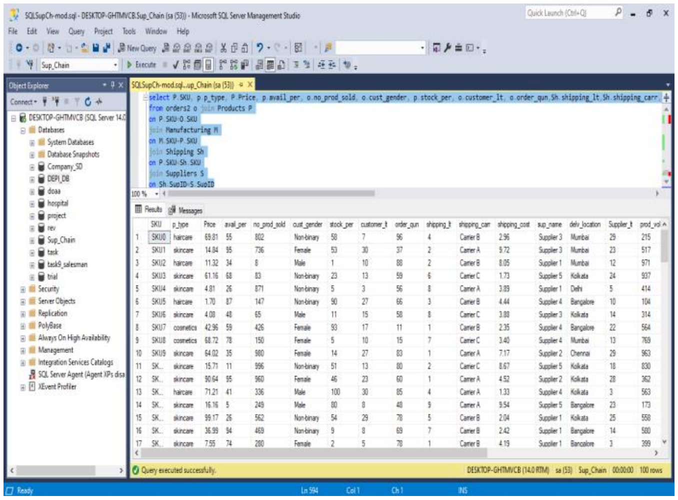
The relationships are:

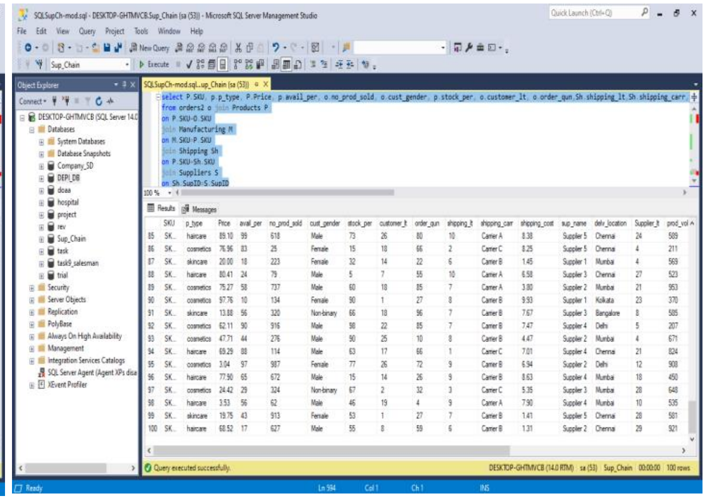
- Products (one to many) with tables (orders - manufacturing – shipping)
- Suppliers (one to many) with tables (manufacturing – shipping)

Diagram of relationships

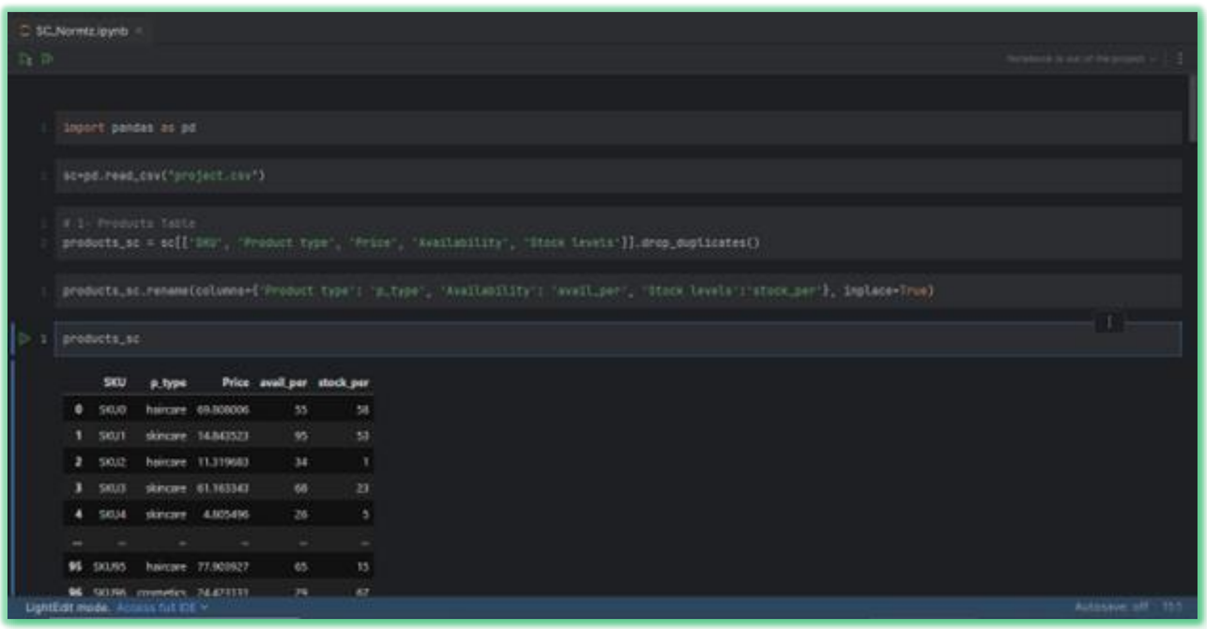


The final table of merging 5 tables:





Normalization code in Python



SC_Normlz.ipynb

```python
products_sc.to_csv("products.csv", index=False)
```

```python
# 2- Orders2 Table
orders_sc = sc[['SKU','Number of products sold','Order quantities','Lead times','Customer demographics']].copy()
```

```python
orders_sc['OrderID'] = orders_sc.index
```

```python
orders_sc.rename(columns={'Order quantities':'order_qun','Number of products sold':'no_prod_sold',
                          'Lead times':'customer_lt','Customer demographics':'gender', 'OrderID':'ord_ID'}, inplace=True)
```

```python
orders_sc
```

	SKU	no_prod_sold	order_qun	customer_lt	gender	ord_ID
0	SKU0	802	96	7	Non-binary	0
1	SKU1	736	37	30	Female	1
2	SKU2	8	88	10	Male	2
3	SKU3	83	59	13	Non-binary	3
4	SKU4	871	56	3	Non-binary	4
...	...	...	...	...	...	...
95	SKU95	672	26	14	Male	95
96	SKU96	324	32	2	Non-binary	96

Analysis Questions

4-Analysis Questions

As mentioned in introduction, Supply Chain has many KPI's that can evaluate its performance like (Perfect Order Rate - Fill Rate - Inventory Turnover Rate ….etc.), but to calculate these KPI's, some variables should be exist in dataset such as: Order Date, Period of data, completed orders, returns, errors, beginning inventory, ending inventory. Unfortunately, our dataset doesn't have any of these variables.

Accordingly, we suppose that we can measure the efficiency of the supply chain by covering three main areas (Efficiency of Inventory Management - Evaluation of Suppling and Shipping - Financial) based on **the main goal of supply chain**:

- **Efficiency of Inventory Management service level**: (stock – availability)
- **Evaluation of Suppling and Shipping**: Time (the Lead Time), Costs (Total Costs - shipping costs) – Quality (Defect Rate, Inspection Result)
- **Financial**: Revenue

Based on these areas, we indicate some questions to be answered by the analysis:

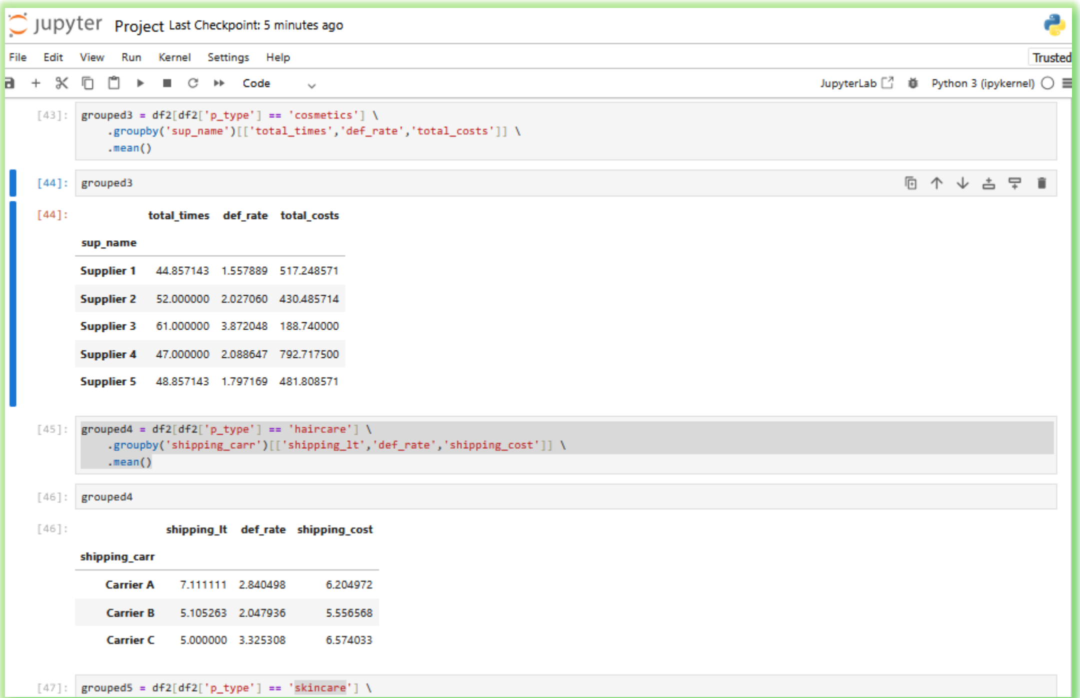
1- Which product type offers the best inventory management and the strongest financial performance?
2- Who tends to purchase more: males or females?
3- Which supplier demonstrates the best overall performance?
4- Which Carrier provides the most reliable service?
5- Which Transportation mode is the most efficient?
6- Which route provides the best overall performance?

In medical/beauty products, high defect rates are due to health and safety risks. So that the main target to defect rate is to be zero. Acceptable Quality Limit (AQL) is a standard measure in this industry that is usually used to evaluate the quality of products. According to AQL, there are three levels of defect rate:

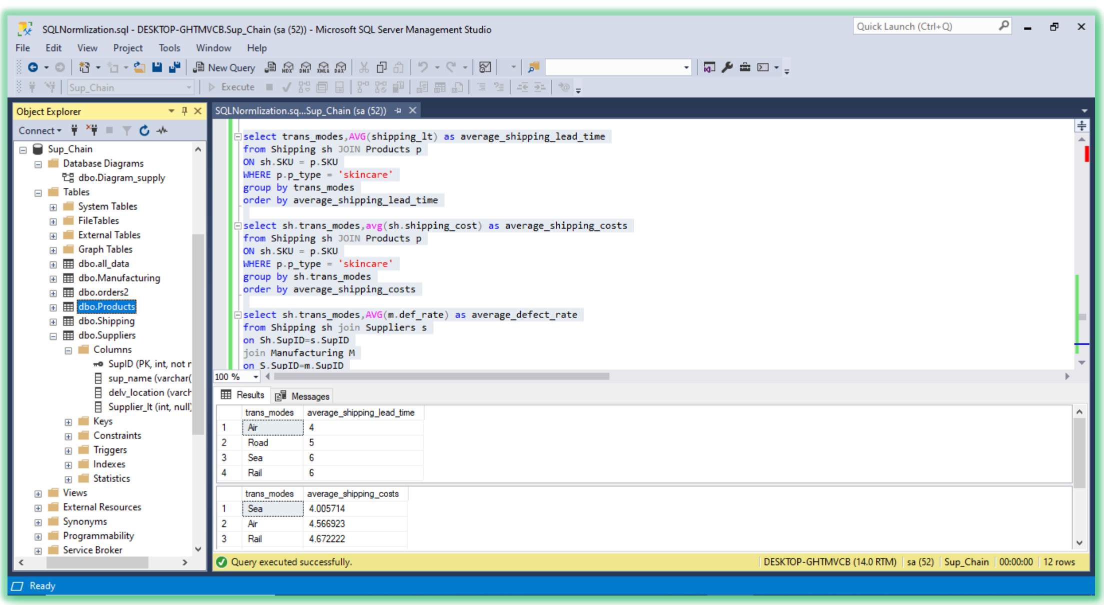
1- Excellent level, if defect rate <1,
2- Acceptable level, if defect rate between (1-2),
3- Actions required, if defect rate >2

As a result, our main concern in supply chain is to use the best ways to decrease the defect rate even with high costs. Low quality in health or beauty industry means increasing costs because of the costs of rework, replacements, and compensation.

Answering questions using Python



Answering questions using SQL



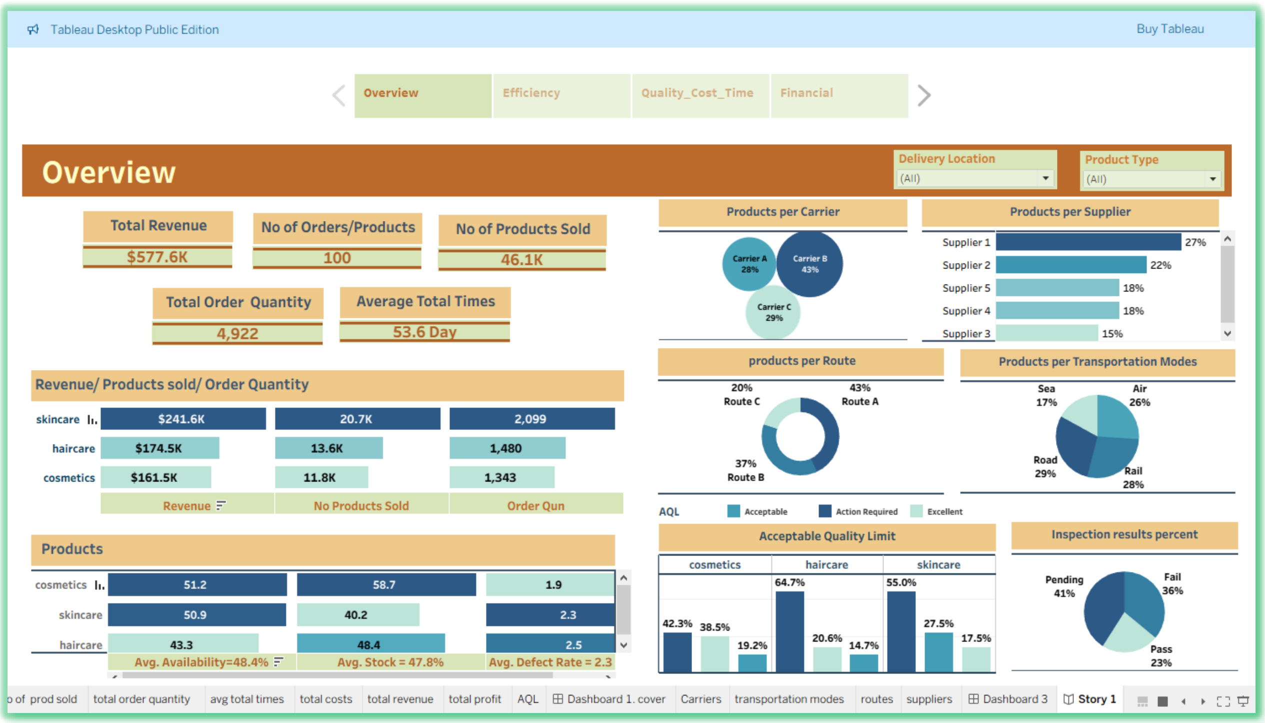
Visualization

Using Tableau, 4 dashboards are designed to answer the analysis questions.

Dashboard 1:

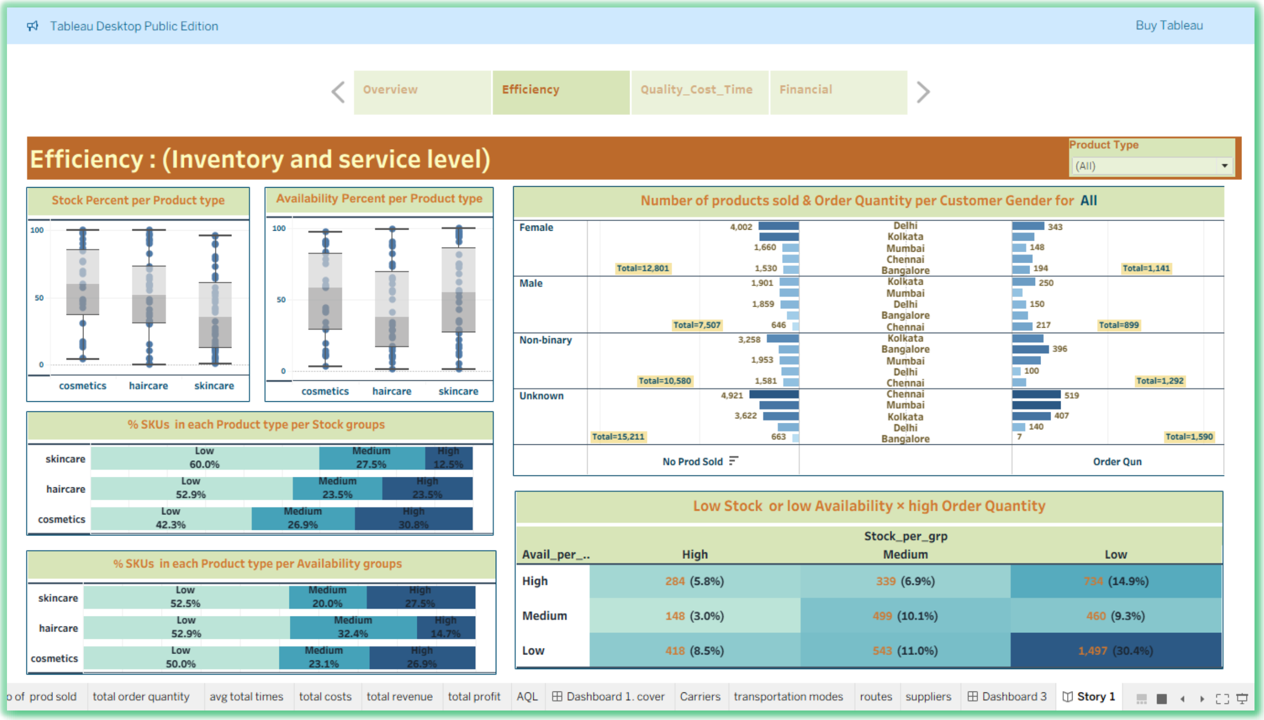
It gives an overview of the whole data. It tells us who makes more orders (Supplier 1 - Carrier B – Transportation Rail – Route A) and Skincare has the most orders and revenue.

Also, Kolkata is the highest-demand delivery location for skincare and haircare products, while Delhi leads in demand for cosmetic products. This indicates that customer preferences for specific product categories are influenced by location.



Dashboard 2:

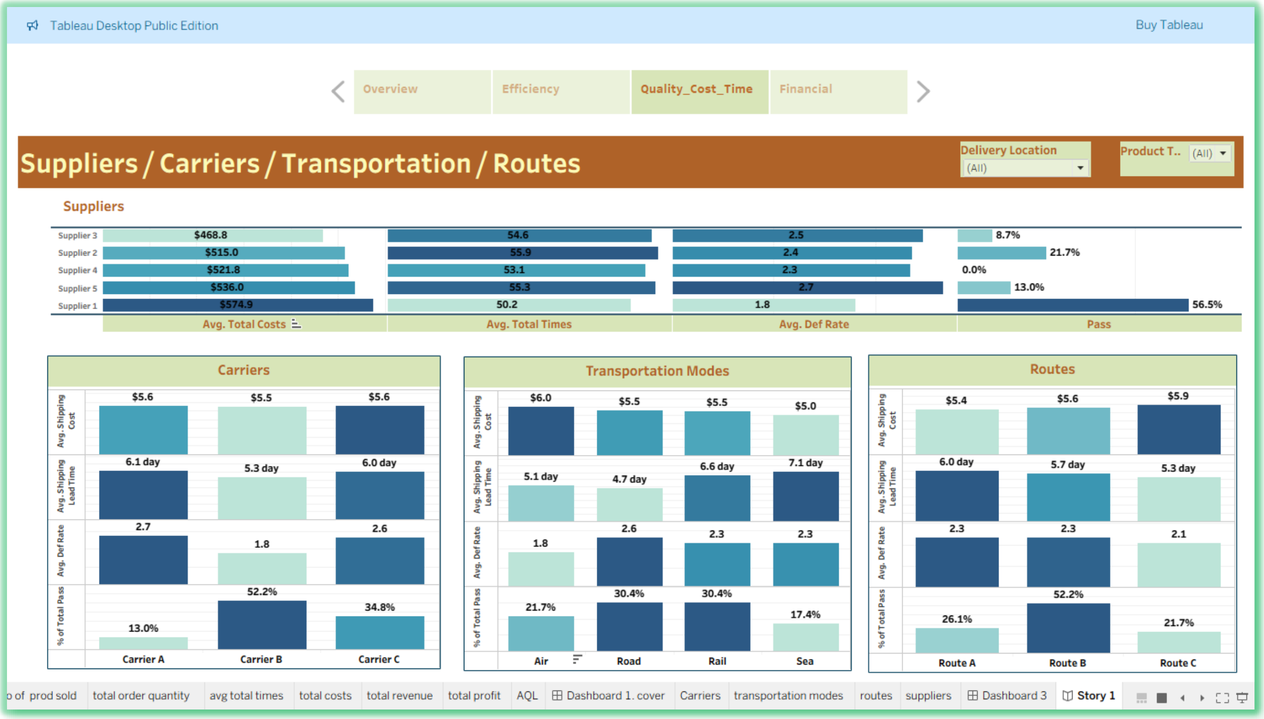
It indicates that although skincare products have the lowest stock percentage compared to haircare and cosmetics, it generates the largest volume of orders and highest revenue. The combination of high demand and low stock percentage may increase the risk of lost sales and customer dissatisfaction if not closely monitored. So that, decision-makers should prioritize high availability of skincare. Also, the dashboard shows that the Unknown gender group has the highest demand, reflected by the largest order quantity and the highest number of products sold. Then, Female and Non-Binary customers follow in demand.



Dashboard 3:

It tells us that:

- For **Cosmetics**: the best is (Supplier 1 -Carrier B – Transportation Air – Route C)
- For **Skincare**: the best is (Supplier 1 -Carrier B – Transportation Air – Route B)
- For **Haircare**: the best is (Supplier 3 -Carrier B – Transportation Rail – Route C)



- **Dashboard 4:**

It shows that:

- Supplier 1 leads to highest revenue in skincare, Supplier 2 in haircare, Supplier 5 in cosmetics
- Carrier B and Route A drive the highest revenue
- Mumbai and Kolkata generate the most revenue; Kolkata excels in skincare, Mumbai in cosmetics
- Unknown customers drive the highest overall revenue, with females leading in skincare and cosmetics, and Unknown leading in haircare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Overview
Efficiency
Quality_Cost_Time
Financial
Financial (Revenue)
Delivery Location
(All)
Product Type
(All)
Revenue per Delivery Location
Mumbai $137.8K (23.8%)
Bangalore $102.6K
Chennai $119.1K
Kolkata $137.1K (23.7%)
Delhi $81.0K
Revenue per Customer Gender
Unknown $173.1K
Female $161.5K
Male $126.6K
Non-binary $116.4K
Total Revenue per Transportation Mode
Rail $165.0K
Road $159.3K
Air $155.7K
Sea $97.6K
Total Revenue per Routes
Route C 20.8%
Route A 43.8%
Route B 35.4%
Total Revenue per Shipping Carrier
Carrier A 24.7%
Carrier B 43.3%
Carrier C 32.0%
Total Revenue per Supplier
$157.5K $125.5K $110.3K $97.8K $86.5K
Supplier 1 Supplier 2 Supplier 5 Supplier 3 Supplier 4
o of prod sold
total order quantity
avg total times
total costs
total revenue
total profit
AQL
Dashboard 1. cover
Carriers
transportation modes
routes
suppliers
Dashboard 3
Story 1

Conclusions

- According to Acceptable Quality Limit (AQL) standard (defect rate <1 is excellent level – 1-2 acceptable level -and >2 actions required). As a result, the Average defect rate for all product types (cosmetics 1.9 – haircare 2.5 – skincare 2.3) needs required actions (cosmetics lies at the end of acceptable level). In medical/personal care products industry, low quality leads to health and safety risks. In our dataset the percent of products that have Defect rate >2 is (cosmetics 42% – haircare 65% – skincare 55%). That is the main reason for low Pass percent (23%)

- For Cosmetics: best (Supplier 1 -Carrier B – Transportation Air – Route C) – for Skincare: best (Supplier 1 -Carrier B – Transportation Air – Route B) – for Haircare: best (Supplier 3 -Carrier B – Transportation Rail – Route C)

- Skincare is the best product type. Although lowest stock level, highest Pass 28%, highest orders, highest revenue

- Kolkata is highest demand for skincare and haircare, while Delhi is for cosmetics

Recommendations

By focusing on these key areas, companies can significantly reduce defects, minimize waste, and improve the overall pass percentage in the supply chain of personal care products:

- Regular Audits and Monitoring: on-site audits, products consistent to agreed-upon quality standards
- Cost Efficiency: "upfront quality" during manufacturing and sourcing reduces costs associated with rework, replacements, and warranty claims later in the supply chain.
- Suppliers: select based on reputation, certifications (ISO) and continuing on-site visits
- Warehousing: closer to key markets, reducing shipping costs and lead times
- Use Technology for Tracking: Implement a Management System to track products throughout the supply chain
- Ensure top-selling products are consistently in stock with rapid replenishment
- Reducing the unknown category in customer gender can be eliminated by validation rule in database before entering

For this dataset, we can recommend decision makers to use:

- For Cosmetics: best (Supplier 1 -Carrier B – Transportation Air – Route C) – for Skincare: best (Supplier 1 -Carrier B – Transportation Air – Route B) – for Haircare: best (Supplier 3 -Carrier B – Transportation Rail – Route C)
- Stock of Skincare should be managed to cover highest orders
- Ensure high-demand locations like Mumbai and Kolkata are well-stocked, particularly Kolkata for skincare and Mumbai for cosmetics
- Female customers order skincare and cosmetics products more than haircare, while males order haircare products more